Palo Alto, CA 94306 **Stephan Popescu** (914)-574-9417
https://www.linkedin.com/in/stephan-popescu-/ snp7432@gmail.com

Professional Experience

Archer Aviation, San Jose CA

Senior Dimensional Engineer *Jan 2026 – Present*

- Leading dimensional engineering for cabin & closure systems on the Midnight aircraft program with a focus on high-volume design practices applied to traditional aerospace composite & metallic structures.
- Developed interactive database for perceived quality, datum strategy, assembly sequence, variation analysis, and inspection planning to improve cross-functional communication.
- Performed 3D tolerance analysis & datum optimization to achieve assembly accuracy targets using 3DCS.

Lucid Motors, Newark CA

Senior Mechanical Engineer – Dimensional Engineering *May 2023 – Dec 2025*

- 3D variation modeling to validate & improve critical 50 functional interfaces between transmission, inverter, and thermal components inside the Atlas Drive Unit with key focus on automation/robotic processes.
- Led system integration development for BIW, Closures, & Exteriors on the Midsize program with a statistical & first principles approach to improve dimensional accuracy.
- Analysis & design input for ADAS hardware, structures, & A-surface trim on the Gravity Robo-taxi program.
- 3D variation modeling/Monte Carlo analysis to validate 100+ functional & craftsmanship targets using 3DCS for all interior components on the Gravity program, enabling early design changes and reducing launch risk.
- BIW/closures fixture conceptual design and tolerance negotiations with tier 1 suppliers.
- Metrology hand measurements techniques, 3D laser scanning, inspection & data analysis using Polyworks.

Mechanical Engineer – Vehicle Architecture *January – June 2023*

- Architectural design of front-end components including BIW, Chassis, Interiors, Exteriors, Thermal, and LV/HV on the Gravity program.
- Led cross-functional packaging reviews to present impacts/tradeoffs of exterior washer systems to senior management which reduced overall piece price by $10 & CAP-EX by $400K.

Milwaukee Tool, Milwaukee WI *June - August 2022*

Advanced Engineering Intern – Mechanics Hand Tools

- Led the development of an industry leading low reverse torque ratcheting mechanism (**US20240082990A1**)
- Benchtop torque to failure testing of prototype parts to gain confidence on initial design path.
- Benchmarking & field research from 8 commercial service professionals to develop a list of key user needs.

Herman Miller, Holland MI - *Product Development Engineering Intern – Seating* *May - August 2021*

Subaru Of Indiana Automotive, Lafayette IN - *Advanced Manufacturing Engineering Intern* *June - August 2020*

GE Aviation, Lynn MA - *Manufacturing/Quality Engineering Co-Op* *January - May 2020*

HDR Engineering., Newark NJ - *Mechanical Engineering Intern* *May - August 2019*

Education

Purdue University, West Lafayette IN *December 2022*

B.S. Mechanical Engineering Technology GPA: 3.66/4.0

Purdue Electric FSAE - *Vehicle Dynamics Team Lead* *August 2021 - July 2022*

- Led a team of 15+ to design, build, & test components of steering, braking, & suspension systems.

Technical Skills

3DX, NX, Creo, Catia V5, SolidWorks
Ansys Mechanical FEA
3DCS Variation Analysis, Polyworks
Fusion 360 CAM (Lathe, 5-Axis Mill)
ASME Y14.5 GD&T, DFM, DFA
Six Sigma processes (RCCA, 5 whys, AWO's)
Jira project & resource management
Stamping, casting, injection molding processes